

Fundamentos de Programación

Grado en Ingeniería del Software

15/09/25

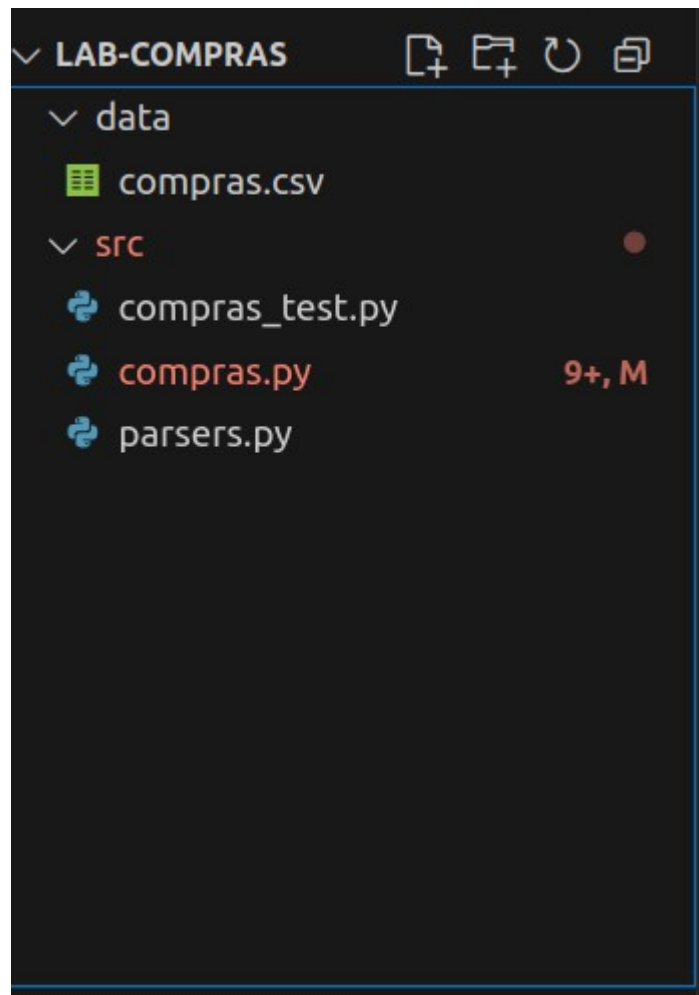
Antecedentes

- Presentación FP
 - Organización
 - Evaluación
- Uso básico de notebooks
- Uso básico de Visual Code Studio

Índice

- Estructura básica programa Python
- Uso de bucles e instrucciones condicionales
- Esquema de transformación (map)

ESTRUCTURA BÁSICA PROGRAMA PYTHON



```
compras_test.py •
src > compras_test.py > ...
1  # -*- coding: utf-8 -*-
2  """
3  Created on Fri Jan 10 09:51:21 2020
4
5  @author: Damián Fernández
6  @revisor: Pepe Riquelme, Mariano González
7  ÚLTIMA MODIFICACIÓN: 24/01/2020
8  """
9
10 from compras import *
11
12 def test_lee_compras(compras):
13     print("Número de registros leídos:", len(compras))
14     print("Tres primeros registros:", compras[:3])
15     print("Tres últimos registros:", compras[-2:])
16
17
18 if __name__ == "__main__":
19     print("\nEJERCICIO 1")
20     compras = lee_compras("data/compras.csv")
21     test_lee_compras(compras)
22
```

```
src > compras.py > lee_compras
```

```
49         return res
```

USO DE BUCLES E INSTRUCCIONES CONDICIONALES

The if-else Statement

```
age = 16
if age < 18:
    print("You can't vote!")
else:
    print("You can vote!")
print("Done!")
```

Short Hand
if-else

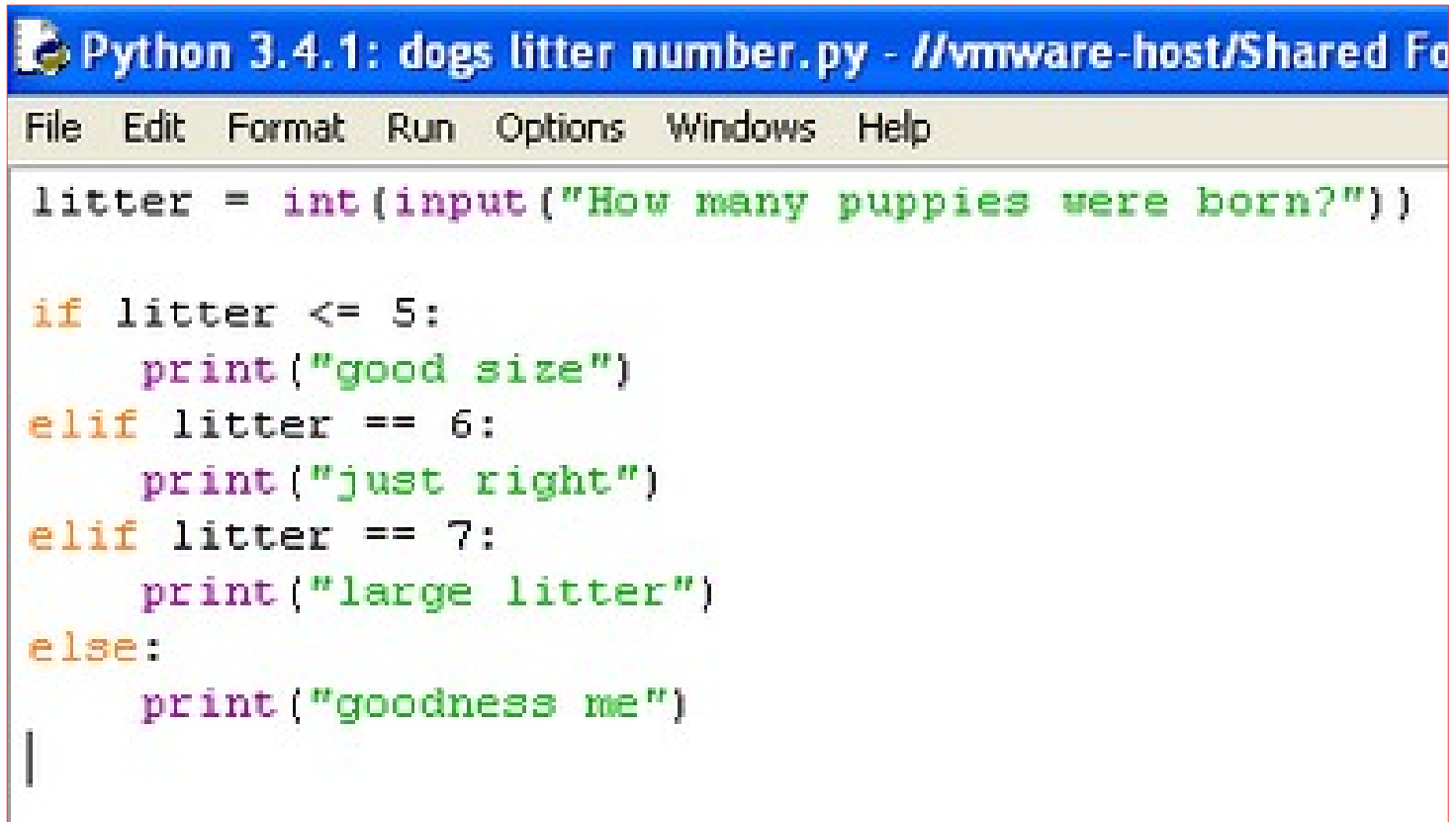


```
age = 20
print("You can't vote!") if age < 18 else print("You can vote!")
print("Done!")
```

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Python

<https://www.youtube.com/watch?v=SiECEKZDqqk>

A screenshot of a Python 3.4.1 IDE window titled "Python 3.4.1: dogs litter number.py - //vmware-host/Shared Fo". The window has a menu bar with "File", "Edit", "Format", "Run", "Options", "Windows", and "Help". The code editor contains a Python script that asks the user for the number of puppies born and prints a response based on the count. The code is as follows:

```
litter = int(input("How many puppies were born?"))

if litter <= 5:
    print("good size")
elif litter == 6:
    print("just right")
elif litter == 7:
    print("large litter")
else:
    print("goodness me")
|
```

Bucle for y la función range()



```
for i in range(1, 10, 3):  
    print (i)
```

A black circular arrow icon, also known as a refresh or loop symbol, is positioned to the right of the code snippet, indicating the iterative nature of the 'for' loop.

<https://www.youtube.com/watch?v=AEYxklU1ZWI>

script.py

```
a_list = [1, 3, 5]
```

```
    iteration variable
      ↓
for value in a_list:
    print(value)
    print(value - 1) | the body of the loop
    iterable variable
      ↓
```

Output

```
1
0
3
2
5
4
```

<https://www.dataquest.io/blog/python-list-tutorial/>

ESQUEMA DE TRANSFORMACIÓN

```
def transforma_cuadrado(lista):  
    """  
    Recibe una lista de enteros y devuelve una nueva lista  
    con cada número elevado al cuadrado.  
  
    Ejemplo:  
    Entrada: [1, 2, 3, 4]  
    Salida: [1, 4, 9, 16]  
    """  
    nueva_lista = []  
    for numero in lista:  
        nuevo = numero ** 2 # aquí hacemos la transformación  
        nueva_lista.append(nuevo)  
    return nueva_lista
```

Python map() function

Last Updated : 07 Sep, 2025



map() function in Python applies a given function to each element of an iterable (list, tuple, set, etc.) and returns a map object (iterator). It is a higher-order function used for uniform element-wise transformations, enabling concise and efficient code.

Let's start with a simple example of using **map()** to convert a list of strings into a list of integers.

```
s = [1, 2, 3, 4]
res = map(lambda x:x*x, s)
print(list(res))
```



```
[1, 4, 9, 16]
```

<https://www.geeksforgeeks.org/python/python-map-function/>

TAREA

- Completar notebooks 1 y 2 de teoría
- Completar ejercicios pendientes de clase.